6.5 First fractions: answers

← Unit overview · Practice

For parents: equivalent answers are correct. If he writes 2/4 where the answer says 1/2, praise it and ask “Is there another name for that?”

Set 1 · Warm-up

#	Answer	How
1	1/2	2 equal parts, 1 shaded.
2	1/4	4 equal parts, 1 shaded.
3	3/4	4 equal parts, 3 shaded.
4	2/3	3 equal parts, 2 shaded.
5	4/8 (= 1/2)	8 equal parts, 4 shaded. That’s half the strip.
6	5	10 shared into 2 equal groups.
7	10	20 shared into 2.
8	4	Four quarters (*vier kwart*) make one whole.
9	2	Two halves make one whole.
10	4	8 shared into 2.

Set 2 · Make a whole

#	Answer	How
1	3/4	4 quarters in a whole; 1 there, 3 missing.
2	3/5	5 fifths in a whole; 2 there, 3 missing.
3	2/3	3 thirds in a whole.
4	3/8	8 eighths in a whole.
5	3/10	10 tenths in a whole.
6	1/2 (or 2/4, 4/8…)	Two halves make a whole.
7	5/8	8 slices, 3 eaten, 5 left.
8	7/12	12 squares, 5 eaten, 7 left.
9	1	Two halves make one whole.

Set 3 · Which is bigger?

#	Answer	How
1	1/2	Shared by 2 gives bigger pieces than shared by 4.
2	1/3	Thirds are bigger than fifths.
3	3/4	Same size pieces (quarters); 3 is more than 1.
4	4/5	Same size pieces (fifths); 4 is more than 2.
5	1/6	Sixths are bigger than eighths.
6	= (the same)	Two quarters fill exactly the same as one half.

#	Answer	How
7	5/6	5/6 is more than half (half would be 3/6).
8	Lina	1/3 is bigger than 1/4.
9	1	9/10 is one tenth short of a whole.
10	= (the same)	Two halves make one whole.

Talk about it: when the top number is 1, the **bigger** the bottom number, the **smaller** the piece, because the whole is shared between more.

Set 4 · The number line

#	Answer	How
1	A = 1/4, B = 1/2 (or 2/4), C = 3/4	The line from 0 to 1 is split into 4 equal steps.
2	D = 1, E = 1 1/2	The line from 0 to 2 is split into 4 steps, so each step is 1/2. D is 2 steps, E is 3 steps.
3	1/10	1/10 is just one small step from 0; 9/10 is close to 1.
4	More	Half of 5 fifths would be 2 1/2 fifths; 3 fifths is more.
5	1/8 < 1/4 < 1/2	Smaller pieces first.
6	1/4	Half of a half is a quarter.
7	1/3 a third of the way along, 2/3 two thirds of the way	Three equal steps.
8	In the middle, between 1/3 and 2/3	Half is a bit more than 1/3 and a bit less than 2/3.

Set 5 · A fraction of an amount

#	Answer	How
1	6	12 : 2.
2	5	20 : 4.
3	5	15 : 3.
4	6	30 : 5.
5	15	1/4 of 20 = 5; 3 × 5 = 15.
6	8	1/3 of 12 = 4; 2 × 4 = 8.
7	€5	€50 : 10.
8	6 children	24 : 4.
9	45 minutes	1/4 of 60 = 15; 3 × 15 = 45.
10	24 sweets	4 quarters in the whole bag: 4 × 6.
11	27 children	3 thirds in the whole class: 3 × 9.
12	50	100 : 2.

Puzzle corner

- **Three unit fractions that make 1:** 1/3 + 1/3 + 1/3; 1/2 + 1/4 + 1/4; 1/2 + 1/3 + 1/6. These are the only three ways with three unit fractions.
- **Different unit fractions:** 1/2 + 1/3 + 1/6 = 1. With four: 1/2 + 1/4 + 1/6 + 1/12 = 1, or 1/2 + 1/4 + 1/5 + 1/20 = 1.
- **Big challenge:** there are infinitely many ways once you allow more fractions. (Any 1/n can be split into 1/(n+1) + 1/(n × (n+1)), for example 1/6 = 1/7 + 1/42.) That's far beyond groep 6, but if he notices "I can keep going", that's real mathematical thinking.

The strips can check 1/2 + 1/3 + 1/6 directly: lay a half, a third and a sixth end to end under the whole strip.